

Varun Budati

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EDUCATION

Virginia Tech , Blacksburg, Virginia	Aug 2023 – May 2027	CFA Level 1 Candidate
B.S. in Computer Science	GPA: 3.59/4.0	
Minor in Mathematics & Finance	In-Major GPA: 3.76/4.0	

SKILLS, COURSEWORK & CERTIFICATIONS

Programming Languages: Python, SQL, JavaScript, Java, C, HTML/CSS

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Plotly, Sklearn, Seaborn, SciPy, Statsmodels, React, Node.js, Flask

Developer Tools & OS: Git, Docker, AWS, Linux/Unix

Certifications: Financial Analysis (Power BI), Akuna Capital Options 201 (ID: 92400251)

WORK EXPERIENCE

Quantitative Research, Dataism Lab for Quantitative Finance - Virginia Tech, Blacksburg, Virginia October 2024 – Present

- Analyzed Bitcoin's market microstructure to develop and implement optimal order execution strategies.
- Created a data-driven market simulator that successfully trained an agent on a 3-day historical Bitcoin dataset, demonstrating the model's ability to learn and exploit short-term liquidity patterns.
- Developed a Deep Reinforcement Learning agent in PyTorch, implementing Double Dueling QDN with Prioritized Replay to learn optimal execution policies from high-frequency data.

Summer Research Intern, MAOP - Virginia Tech, Blacksburg, Virginia May 2025 – July 2025

- Analyzed 24,500+ words from intro CS courses at MIT, Berkeley, and CalTech to find pedagogical gaps.
- Developed a coding framework and used text analysis tools (LIWC-22) to thematically categorize examples.
- Provided data-driven recommendations to enhance CS curricula with inclusive, real-world applications.

Research Mentor & Assistant, REACH Lab - Virginia Tech, Blacksburg, Virginia March 2024 – June 2025

- Conducted an extensive literature review on Rural Computer Science Education, synthesizing over 80+ articles and papers to support a \$500,000 National Science Foundation (NSF) grant proposal.
- Mentored and trained 6 research assistants in advanced literature review methodologies and database search techniques.

PROJECT WORK

Sports Algorithm & Analytics System August 2024 – May 2025

- Engineered a quantitative sports prediction model using Python (NumPy, Pandas) to identify statistical edges, achieving an 8,400% return on investment (scaled from \$10 to \$850) over a 6-month period.
- Built a real-time data pipeline with sports APIs to process player stats for probability calculations (SciPy).
- Developed a performance dashboard to visually present key analytics and an automated risk system using the Kelly Criterion for bankroll optimization.

Momentum Trading Strategy Development May 2024 - August 2024

- Independently designed and developed a complete momentum trading strategy in Python, from implementing technical indicators (MACD, RSI) to generating automated signals.
- Backtested the strategy against a benchmark, analyzing performance and visualizing results with Pandas/Matplotlib.

Poker Game January 2024 - March 2024

- Engineered a card handling system using JavaScript arrays and objects to manage a 52-card deck with 4 suits and 13 values.
- Programmed 8 distinct game variants through conditional logic and dynamic payable adjustments.
- Architected poker hand evaluation algorithms with $O(n \log n)$ time complexity for pattern recognition (pairs, straights, flushes).

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Treasurer, FinTech Club, Virginia Tech, Blacksburg, Virginia October 2024 - Present

- Manage the club's budget and all financial operations with detailed record-keeping to fund initiatives and workshops for a community of 100+ members.
- Organize speaker events with industry professionals to create networking and career development opportunities.
- Led a faculty-advised research project replicating the foundational Evans & Archer (1968) paper on portfolio diversification.
- Modeled risk vs. portfolio size (Pandas, NumPy), confirming unsystematic risk is mitigated with 10-20 assets.

IMC Prosperity 3 | Top 0.005% April 2025

- Placed 15th in the US out of 12,600 teams in IMC's Prosperity 3 trading competition, designing strategies for multi-asset markets.
- Optimized trade execution constraints by using fair value estimators (VWAP, EMA, stochastic modeling).
- Designed signal-driven market-making strategies by analyzing mean-reversion patterns and synthetic mispricing via EDA (rolling z-scores, spread compression, correlation clustering).